XUE PAN, Ph.D. *Curriculum Vitae*

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EDUCATION

- **Ph.D. Oilseed Biotechnology** 2010 – 2015
  University of Alberta, Canada
  Adviser: Prof. Randall Weselake

- **M.Sc. Agronomy and Physiology of Oilseed crops** 2007 – 2010
  Dalhousie University, Canada
  Adviser: Prof. Claude Caldwell

- **B.Sc. Horticulture.** 2003 – 2007
  Fujian agriculture and Forestry University, China

PROFESSIONAL POSITIONS

- **Assistant Professor** 2022– present
  University of Toronto, Scarborough, Canada

- **Assistant Project Scientist** 2020 – 2022
  University of California, Riverside, United States

- **NSERC Postdoctoral Fellow** 2016 – 2019
  University of California, Riverside, United States

- **Part-time Adjunct Faculty** 2017 – 2018
  San Bernardino Valley College

AWARDS

- **Natural Sciences and Engineering Research Council of Canada (NSERC) postdoctoral fellowship** (Rank 2nd in Cellular and Molecular Biology Category), April 2016 – March 2018
- **Recipient of the Alberta Innovates Graduate Student Scholarship**, May 2013 – April 2015
- Outstanding Graduate Student Research Award, University of Alberta, Nov. 2015
- J Gordin Kaplan Graduate Student Award, University of Alberta, May 2015
- 1st place award in the 2015 Biotechnology Division Student Paper Competition at 106th AOCS annual meeting and industrial showcases, May 2015
- Oral Presentation Runner-up Award at 21st International Symposium on Plant Lipids, July 2014
- Student Travel Award at 21st International Symposium on Plant Lipids, July 2014
- Student Achievement Award from the AOCS Biotechnology Division (1st place), May 2012

- Entrance Scholarship from University of Alberta, Sept. 2010
- Plant Science Entrance Scholarship from Dalhousie University, Sept. 2007

PUBLICATIONS IN REVIEW

1. Huang, A., Sun, W., Lv, K., Sang, Y., **Pan, X.,** Fu, Y., Liu, B., Liu, R., Yang, Z. Natural variation of OsRopGAP5 is associated with geographic distribution of rice. *The Plant Cell.* Submitted.

PEER-REVIEWED PUBLICATIONS

1. Rong, D., Zhao, S., He, H., Wang, Z., Ma, H., Huang, Y., Yao, X., Liu, R., **Pan, X.,** Lv, L., Xiao, J., Liu, R., Nagawa, S., Yamamuro, C. A ROP signaling pathway regulates spatial pattern of cell division and specification of meristem notch. *PNAS*.

2. Chen, B., Dang, X., Bai, W., Zhu, L., Liu, X., Yang, Y., Yu, P., Chen, Y., Wang, H., **Pan, X.,** Ren, H, Qin, Y., Wang, Q., Lin, D. (2022) The IPGA1-ANGUSTIFOLIA module regulates microtubule organization and pavement cell shape in Arabidopsis. *New Phytologist.* Link

3. Lin, W., Tang, W., **Pan, X.,** Huang, A., Anderson, C.T., Yang, Z. (2022) Arabidopsis pavement cell morphogenesis requires FERONIA binding to pectin for activation of ROP GTPase signaling. *Current Biology*. 32, 497-507. Link
recommended in F1000Prime as being of special significance in its field

4. Lin, W., Zhou, X., Tang, W., Takahashi, K., **Pan, X.**, Dai, J., Ren, H., Zhu, X., Pan, S., Zheng, H., Gray, W.M., Xu, T., Kinoshita, T., Yang, Z. (2021) TMK-based cell surface auxin signaling activates cell wall acidification in Arabidopsis. *Nature*. 599, 278–282. Link
recommended in F1000Prime as being of special significance in its field

5. Xu, Y[*]., **Pan, X.[*],** Lu, J., Wang, J., Stout, J., Chen, G. (2021) Evolutionary and biochemical characterization of a Chromochloris zofingiensis MBOAT with wax synthase and diacylglycerol acyltransferase activity. *Journal of Experimental Botany*. Link
*co-first authors

6. **Pan, X.,** Fang, L., Liu, J., Senay-Aras, B., Lin, W., Zheng, S., Zhang, T., Manor, U., Chen, W., Yang, Z. (2020) Auxin-induced signaling protein nanoclustering contributes to cell polarity formation. *Nature Communications*. Link

7. Subedi, U., Rathnayaka, K., **Pan, X.,** Ozga, J., Chen, G., Foroud, N., Singer, S.D. (2020) The potential of genome editing for improved seed oil content and composition in oilseed species. *Lipids.* Link

8. Mapelli-Brahm, A., Sanchez, R., **Pan, X.,** Martínez-Force, E., Garces, R., Moreno-Pérez, A.J., Weselake, R.J., Salas, J.J., Venegas-Calerón, M. (2020) Molecular and functional characterization of LYSOPHOSPHATIDYLCHOLINE: ACYL-COA ACYLTRANSFERASE genes from sunflower (*Helianthus annuus L.). Frontiers in Plant Science.* Link

9. Xu, Y., Holic, R., Li, D., **Pan, X.,** Mietkiewska, E., Chen, G., Ozga, J., Weselake, R.J. (2018) Substrate preferences of long-chain acyl-CoA synthetase and diacylglycerol acyltransferase

contribute to enrichment of flax seed oil with α -linolenic acid. *Biochemical Journal.* 475(8), 1473-1489 Link

10. Mietkiewska, E., **Pan, X.,** Shah, S., Weselake, R.J. (2017) Gene combinations for production of bioactive and industrial oils. *Inform.* 28 (3), 15. Link

11. Greer, M.S., **Pan, X.,** Weselake, R.J. (2016) Two clades of type-1 *Brassica napus* diacylglycerol acyltransferase exhibit differences in acyl-CoA preference. *Lipids*. 51 (6), 781-786. Link

12. **Pan, X.,** Chen, J., and Yang, Z. (2015) Auxin: A trigger for cell polarization and morphogenesis in plants. *Current Opinion in Plant Biology*. 28, 144-153. Link

13. **Pan, X**., Chen, G., Kazachkov, M., Greer, M.S., Caldo, K.M.P., Zou, J. and Weselake, R.J. (2015) *In vivo* and *in vitro* evidence for biochemical coupling of reactions catalyzed by lysophosphatidylcholine acyltransferase and diacylglycerol acyltransferase. *Journal of Biological Chemistry.* 290:18068–18078. Link

14. **Pan, X.**, Peng, F.Y. and Weselake, R.J. (2015) Genome-wide analysis of PHOSPHOLIPID:DIACYLGLYCEROL ACYLTRANSFERASE (PDAT) genes in plants reveals the eudicot-wide PDAT gene expansion and altered selective pressures acting on the core eudicot PDAT paralogs. *Plant Physiology*. 167(3): 887–904. link

15. Chen, G., Woodfield, H.K., **Pan, X.**, Harwood, J.L. and Weselake, R.J. (2015) Acyl editing and control of carbon flow in plant oil accumulation. *Lipids*. 50 (11), 1057-1068. Link

16. Wickramarathna, A.D., Siloto, R.M.P., Mietkiewska, E., Singer, S.D., **Pan, X.** and Weselake, R.J. (2015) Heterologous expression of flax *PHOSPHOLIPID:DIACYLGLYCEROL CHOLINEPHOSPHOTRANSFERASE* (*PDCT*) increases polyunsaturated fatty acid content in yeast and Arabidopsis seeds. *BMC Biotechnology*. 15:63. Link

17. **Pan, X.**, Siloto, R.M.P., Wickramarathna, A.D., Mietkiewska, E. and Weselake, R.J. (2013) Identification of a pair of phospholipid:diacylglycerol acyltransferases from developing flax (*Linum usitatissimum* L.) seed catalyzing the selective production of trilinolenin. *Journal of Biological Chemistry.* 288(33):24173-88. Link

18. **Pan, X.**, Caldwell, C.D., Falk, K.C., and Lada, R. (2012) The effect of cultivar, seeding rate and applied nitrogen on *Brassica carinata* seed yield and quality in contrasting environments. *Canadian Journal of Plant Science.* 92: 961-971. Link

19. **Pan, X.**, Lada, R. Caldwell, C.D., and Falk, K.C. (2011) Water-stress and N-nutrition effects on photosynthesis and growth of *Brassica carinata. Photosynthetica.* 49: 309-315. Link

20. **Pan, X.**, Lada, R. Caldwell, C.D., and Falk, K.C. (2011) Photosynthetic and growth responses of *Camelina sativa* (L.) Crantz to varying nitrogen and soil water status. *Photosynthetica.* 49:316-320. Link

BOOK CHAPTERS

1. Singer, S.D., Greer, M.S., Mietkiewska, E., **Pan, X.,** and Weselake, R.J. (2013). Genetic engineering of lipid biosynthesis in seeds. *Biotechnology of Crucifers.* pp111-149. Link

2. **Pan, X.,** Fang, Z., Lin, Y.,Lai, Z. (2009) Study on the effects of the ratios of ammonium-N/nitrate-N on the proliferation of the tube plantlets in *Kalanchoe blossfeldiana. Plant Cell Tissue Culture Technology.* 145-154. Link

CONFERENCE ABSTRACTS

1. Xu, Y., **Pan, X.**, Lu, J., Wang, J., Shan, Q., Stout, J., Chen, G. (2021) Biochemical characterization of a wax synthase from a green microalga *Chromochloris zofingiensis. 2021 AOCS Annual Meeting (online)*

2. **Pan, X.,** Wickramarathna, A. D., Xu, Y., Mietkiewska, E., Siloto, R. M. P., Chen, G., Holic, R., Kazachkov, M., Greer, M. S., Caldo, K. M. P., Li, D., Singer, S. D., Ozga, J., Zou, J., Weselake, R. J. (2018) Functional analysis of α-18:3-enrichment of flax oil using a yeast system. *Canadian Lipid & Bioresource Conference*. Saskatoon, SK, Canada

3. **Pan, X.**, Peng, F. Y., Weselake, R. J. (2016) Genomic analysis reveals molecular mechanisms underlying enhanced alpha-linolenic acid accumulation in developing flax seed. *Plant and Animal Genome XXIV Conference*. San Diego, CA, USA

4. **Pan, X.**, Stymne, S., Zou, J., Qiu, X., Chen, G., Weselake, R. J. (2015) *In vivo* and *in vitro* evidence for biochemical coupling of reactions catalyzed by lysophosphatidylcholine acyltransferase and diacylglycerol acyltransferase. *106th AOCS Annual Meeting*. Orlando, FL, USA

5. Weselake, R. J., Greer, M. S., **Pan, X.**, Xu, Y., Mietkiewska, E., Chen, G., Singer, S. D. (2015) Insights into the role of acyltransferases in seed oil biosynthesis. *25th Canadian Conference on Fats and Oils*. Quebec City, CANADA

6. **Pan, X.**, Siloto, R. M. P., Weselake, R. J. (2014) Possible mechanisms for channeling of alpha-linolenic acid into flax seed oil. *21st International Symposium on Plant Lipids*. Guelph, CANADA

7. **Pan, X.**, Siloto, R. M. P., Weselake, R. J. (2013) Flax oil assembly characteristics. *Total Utilizing of Flax GENomics (TUFGEN) Wrap-up Meeting*. Saskatoon, SK, CANADA

8. **Pan, X.**, Siloto, R. M. P., Weselake, R. J. (2013) Identification of a flax phospholipid: diacylglycerol acyltransferase (PDAT) with enhanced selectivity for substrates containing α-linolenic acid. *7th Plant Bio-Industrial Oils Workshop*. Saskatoon, SK, CANADA

9. **Pan, X.**, Siloto, R. M. P., Weselake, R. J. (2012) Using molecular genetic strategies to investigate the triacylglycerol biosynthetic pathway in flax. *103rd AOCS Annual Meeting & Expo*. Long Beach, CA, USA,

10. **Pan, X.**, Siloto, R. M. P., Wickramarathna, A. D., Weselake, R. J. (2012) A systematic characterization of triacylglycerol-synthesizing enzymes from flaxseed. *Canadian Society of Plant Biologists Annual General Meeting*. Edmonton, AB, CANADA

11. **Pan, X.**, Siloto, R. M. P., Wickramarathna, A. D., Weselake, R. J. (2011) Triacylglycerol biosynthesis in flax. *Total Utilizing of Flax GENomics (TUFGEN) Annual Meeting*. Saskatoon, SK, CANADA

12. **Pan, X.**, Xie, W., Caldwell, C.D. and Anderson, D.M. (2011). Growth performance and carcass composition of Rainbow trout (*Oncorhynchus mykiss*) fed practical diets containing graded levels of high fat residue Camelina meal. *Canadian Journal of Animal Science.* 91:484.

13. **Pan, X.**, Siloto, R. M. P., Weselake, R. J. (2010) Oil assembly in flax. *Total Utilizing of Flax GENomics (TUFGEN) Annual mMeeting*. Saskatoon, SK, CANADA